MIL-E-1/1543A(USAF)
21 September 1972
SUPERSEDING
MIL-E-1/1543(USAF)
4 JANUARY 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 6166A 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

DESCRIPTION: Tetrode, metal-ceramic

Dimensions: See figure 1

Weight: 15 pounds, nominal

Mounting position --- Vertical, anode up or down

ABSOLUTE RATINGS: Fl = 220 MHz

Parameter: / Unit:	Ef V	Eb kVdc	Ec2 kVdc	Ec1 kVdc	Ib Adc	Ic1 Adc
Maximum:						
B TV:	5.0 ± 5%	7.5	2.0	----	4.0	---
C TV:	5.0 ± 5%	7.5	2.0	-1.0	4.0	---
C Telep:	5.0 ± 5%	5.5	2.0	-1.0	2.0	0.6
C Teleg:	5.0 ± 5%	7.5	2.0	-1.0	3.0	0.6
AB RF SSBSC	5.0 ± 5%	7.5	2.0	----	2.8	0.6
TEST CONDITIONS:	5.0 Vac	5.8	1.2	-200	2.3 to 2.4	---

ABSOLUTE RATINGS: Fl = 220 MHz

Parameter: / Unit:	Pp kW	Pg2i W	Pg1 W	Pi kW	Radiator °C	Mod
Maximum:						
B TV:	12	400	300	24	180	---
C TV:	12	400	300	24	180	Grid
C Telep:	8	270	---	10	180	Anode
C Teleg:	12	400	---	20	180	---
AB RF SSBSC	12	400	---	20	180	----
TEST CONDITIONS:	---	----	---	---	---	----

GENERAL:

Qualification: Not required, substitute First Article Test (See note 13)

1/ Formerly Type 6166A/7007

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
1261	Electrode voltage (1) (grid)	Eb = 1.5 kVdc; ib = 11a; t = 300 See notes 1 and 2	0.65	II	ec1	---	315	v
1256	Electrode current (1) (grid)	Eb = 1.5 kVdc; ib = 11a; See notes 1, 2, and 3	0.65	II	ic1	---	1.5	a
1261	Electrode voltage (2) (grid)	Eb = 1.5 kVdc; Ec1/Ib = 0.02 Adc See note 1	0.65	II	Ec1	---	-225	Vdc
1256	Electrode current (2) (grid)	Eb = 5.0 to 6.4 kVdc Ec1/Ib = 1.65 Adc See notes 4, 5, and 7	0.65	II	Ic1	---	-55	μAdc
1301	Filament current	See note 1	0.65	II	If	165	183	A
----	Operation	Po = 7.0 kW; F = 216 MHz; t = 1,800 See notes 4, 5, and 8	0.65	II	---	---	---	---
1256	Electrode current (3) (grid)	Po = 7.0 kW; F = 216 MHz; See notes 4, 5, and 8	0.65	II	Ic1	0.01	---	Adc
1256	Electrode current (screen)	Po = 7.0 kW; F = 216 MHz; See notes 4, 5, and 8	0.65	II	Ic2	---	0.2	Adc
1231	Emission	F = 216 MHz; See notes 4, 5, 8, and 9	0.65	II	Ef	---	4.5	Vac
----	Hum	Ec1 = -300 Vdc; Ib = 1 Adc; Po = 1.5 kW; F = 216 MHz; t = 300; See notes 4 and 6	0.65	II	HUM	---	-35	db
1331	Direct-interelectrode capacitance	See note 10	0.65	II	Cg1-a	---	0.60	pF
		See note 11			Cg1-f	39.0	47.0	pF
		See note 11			Cg1-g2	52.0	64.0	pF
		See note 11			Cg2-a	21.2	25.8	pF
		See note 10			Ca-f	---	0.080	pF
	Quality conformance inspection, part 3							
----	Service-life guarantee	Po = 7.0 kW; F = 216 MHz; t = 1,000 hours; See notes 4, 5, and 12	---	---	---	---	---	---

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NOTES:

1. The amount of air required through the radiator shall be supplied from a blower equivalent to "Fasco" No. 50749 blower (140 cfm free delivery, 0.9 inch (22.9 mm) blocked static pressure).

2. With the grid of the tube biased to -300 Vdc, the grid is pulsed positively with a 1 μs square wave with a repetition rate of 4,000 pps, to give the specified peak anode current. The peak anode current (ib) is measured on an oscilloscope connected across a calibrated resistor in series with the anode. The peak-grid current is obtained by multiplying the average-grid current by ib/Ib. The average anode current (Ib) is measured by a meter in series with the anode.

3. Read immediately following the 5-minute test time required for electrode voltage (grid) (ec1).

4. The amount of air through the radiator shall be such that the 3.0 inches (76.2 mm) minimum differential water pressure exists across the radiator. Airflow across the seals must keep the seals below 180°C maximum.

5. Test to be made in RCA TT10AH transmitter, or equivalent.

6. Hum-test equipment includes RCA model TT10AH transmitter, or equivalent, RCA model 69C distortion meter, or equivalent, and calibrator circuit (see figure 2).

   Hum test to be made according to the following procedure:

   (a) Adjust black level control to increase bias to specified value.
   (b) Reduce excitation to obtain specified power output and resonate anode tuning.
   (c) Operate for specified time. Readjust transmitter anode tuning.
   (d) Set selector switch on calibrator to "RF" position and read voltage on calibrator meter.
   (e) Set calibrator selector switch to "AC" position and adjust "AC" control to obtain same voltage as read in (d) above.
   (f) Calibrate RCA model 69C distortion meter, or equivalent, as follows:

      (1) Turn "Input" switch to "Check" position.
      (2) Turn all amplitude controls (Coarse, Medium, Fine) to zero.
      (3) Turn "Selector" switch to "Calibrate" position.
      (4) Adjust "zero" control to zero percent distortion as read on meter.
      (5) Turn "Selector" switch to "zero db" noise level.
      (6) Adjust "Gain" control to 10-percent distortion as read on meter.
      (7) Turn "Input" switch to "High Audio" position.
      (8) Adjust amplitude controls (Coarse, Medium, Fine) for zero db meter reading.

   (g) Turn selector switch on calibrator to "RF" position and selector switch on distortion meter to -30 db.
   (h) Hum reading is the sum of selector reading (-30) and distortion meter reading.

7. With the conditions as indicated, record electrode current (1) (grid) as 1Ic1 at the end of specified time. Then set Ec1 = -200 Vdc and record electrode current (1) (grid) as 2Ic1. Electrode current (2) (grid) shall be determined by the following equation: Ic1 = 1Ic1-2Ic1.

8. With the transmitter operated at black level, the tube shall operate at the indicated conditions for the specified time. Electrode current (3) (grid), electrode current (screen), and emission shall be read immediately following the operation test.

9. Reduce filament boltage until power output reading is 500 watts less than that used in operation test, then read filament voltage.

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NOTES: -Continued

10. Use external flat metal shield 12-inch (304.8 mm) square having center hole 4.313(109.55) mm diameter. Shield is located in plane of the grid (2) terminal, perpendicular to the tube axis, and is connected to grid (2). All unused elements grounded.

11. Measured without shield. All unused elements grounded.

12. The tube manufacturer warrants the tube for 1 year, or 1,000 hours of heater life (whichever occurs first). This warranty applies only when the tube is operated within the maximum ratings in 6.5 of MIL-E-1. A defective tube shall either be replaced, or at the option of the contracting officer a credit shall be made in the amount of the original purchase price prorated on the basis of 1,000 hours of "heater-on" time.

13. First Article Testing: The First Article sample furnished by the contractor/manufacturer shall consist of preproduction tubes in the quantity indicated by Table 1 of MIL-E-1. In addition, the contractor shall furnish along with the test reports, written assurance that:

    a. The First Article sample is representative of the product to be furnished under the contract.

    b. Production articles will be fabricated in the facility that produced the sample tubes.

    c. Preproduction and Production articles will be fabricated from new materials (i.e., not used or reconditioned materials or source materials/parts such as Government surplus or reclaimed scrap tubes).

    The contractor shall furnish a test plan to the Contracting Officer for approval before proceeding with First Article testing. The test plan shall list all production test and laboratory test facilities to be utilized in First Article testing and production phases of the contract. Upon receipt of test plan approval, the contractor shall proceed with First Article testing. First Article testing shall include all tests and requirements of this specification and conform to Table 1 of MIL-E-1 where applicable. Upon conclusion of testing, three copies of the test report prepared in accordance with MIL-STD-831 and certified by the Government Representative shall be forwarded along with the test samples to the Contracting Officer for approval.

Custodian:

Air Force - 80

Review activities:

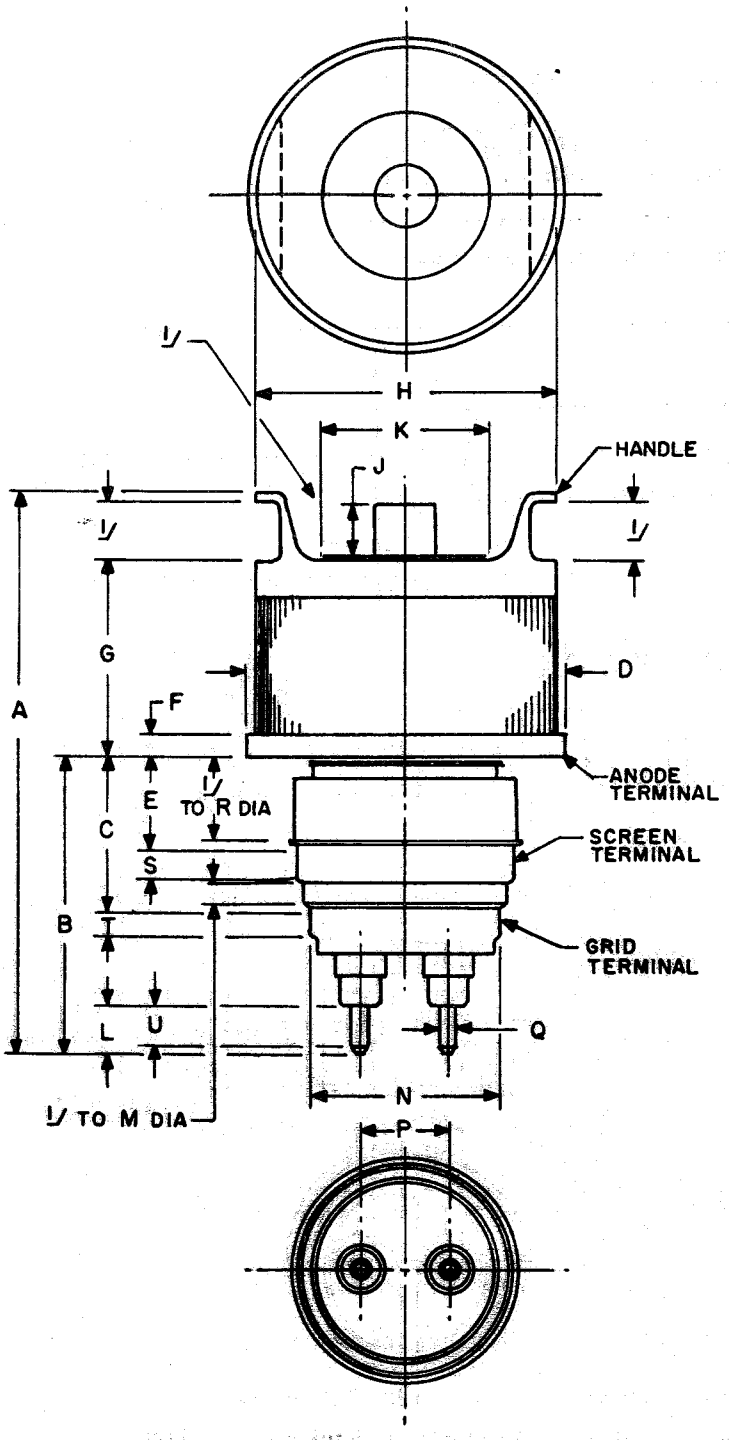
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Preparing activity:

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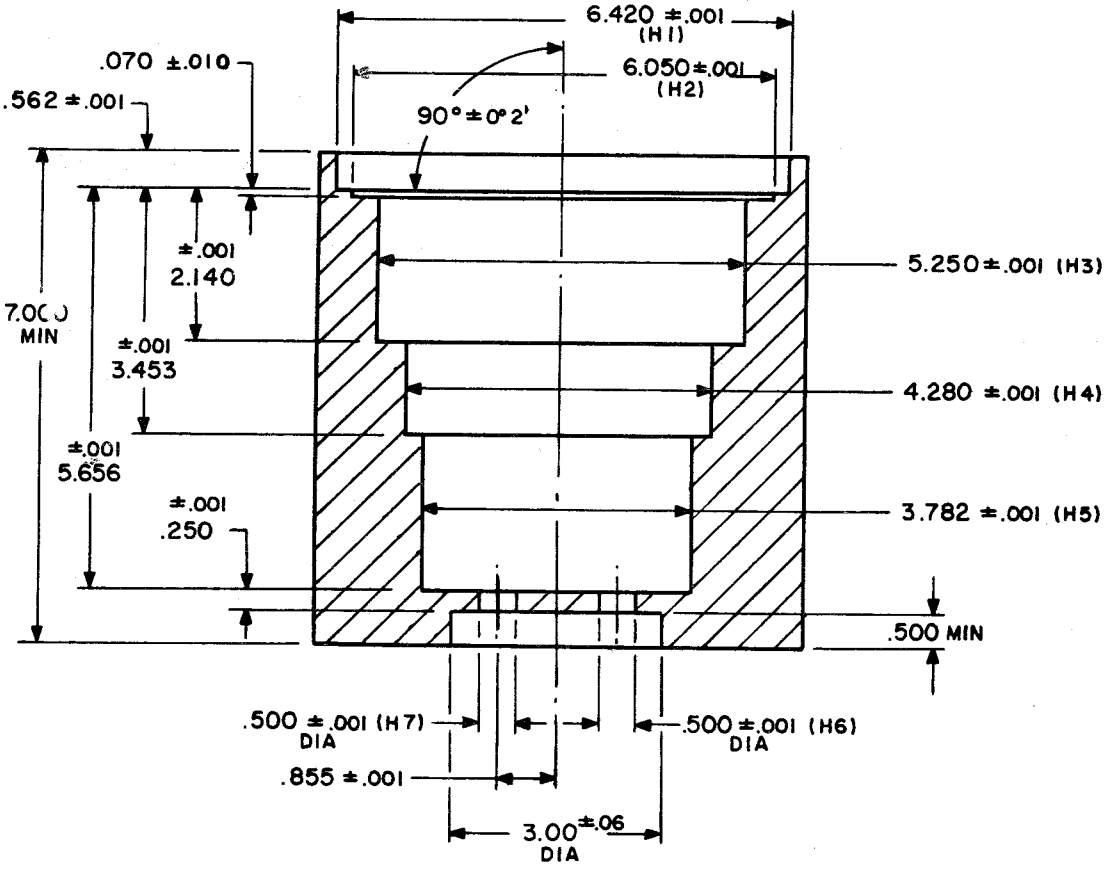


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
QUALITY CONFORMANCE INSPECT., PART 3 (PERIODIC CHECK)					
G	–	4.100	–	104.14	
K	–	3.380	–	85.85	
U	.625	–	15.88	–	
QUALITY CONFORMANCE INSPECT., PART 1					
D	6.340	6.380	161.04	162.05	a
J	–	1.165	–	29.59	b
M	4.226	4.266	107.34	108.36	a,c
N	3.662	3.710	93.01	94.23	a,c
P	1.670	1.750	42.42	44.45	a
Q	.430	.444	10.92	11.28	a
R	–	4.500	–	114.30	a
QUALITY CONFORMANCE INSPECT., PART 2					
A	–	11.500	–	292.10	
B	5.870	6.110	149.10	155.19	
C	3.060	3.180	77.72	80.77	
E	1.780	1.860	45.21	47.24	
F	.470	.530	11.94	13.46	
H	5.900	6.100	149.86	154.94	
L	–	1.093	–	27.76	
S	.500	–	12.70	–	d
T	.350	–	8.89	–	d

1/ NOTE:- KEEP AREAS NOTED CLEAR. DO NOT ALLOW CONTACTS OR CIRCUIT COMPONENTS TO PROTRUDE INTO THESE AREAS.

FIGURE 1. Outline drawing of electron tube type 6166A

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G1 GAGE

NOTES:

a. WITH THE CYLINDRICAL SURFACES OF ITS GRID AND FILAMENT TERMINALS CLEAN, SMOOTH AND FREE OF BURRS, TUBE WILL ENTER A GAGE AS SHOWN IN SKETCH G1. THE FIVE CYLINDRICAL HOLES H1, H2, H3, H4 AND H5 HAVE AXES COINCIDENT WITHIN .001 INCH AND THE AXES SHALL BE PARALLEL TO THE AXES OF H1, H2, H3, H4 AND H5 TO WITHIN 0°2'.
THE ANODE RING WILL BE ENTIRELY ENGAGED BY HOLE H1 AND THE ANODE RING WILL SEAT IN THE SHOULDER BETWEEN H1 AND H2. THE PLANE SURFACE OF THIS SHOULDER IS 90° ± 0°2' TO THE AXES OF THE HOLES. SEATING IS DETERMINED BY FAILURE OF A .020 INCH THICKNESS GAGE TO ENTER MORE THAN .06 INCH BETWEEN SHOULDER SURFACE AND ANODE RING. SLOTS ARE PROVIDED TO PERMIT THIS MEASUREMENT TO BE MADE.

b. WHEN TUBE IS PLACED ON A FLAT SURFACE WITH HEATER TERMINAL UPWARD, IT MUST REST ON THE RADIATOR HANDLES, AND NOT ON THE EXHAUST TUBULATION COVER.

c. THE DIAMETERS M AND N SHOULD BE WITHIN LIMITS THROUGHOUT THE LENGTHS S AND T RESPECTIVELY.

d. THESE DIMENSIONS ARE TO BE RECORDED AS GO OR NO-GO; ACTUAL DIMENSION READINGS ARE NOT REQUIRED.

e. THE AQL FOR DIMENSIONS LISTED UNDER QCI PART I SHALL BE AT INSPECTION LEVEL II.

FIGURE 1 Outline drawing of electron tube type 6166A (Cont)

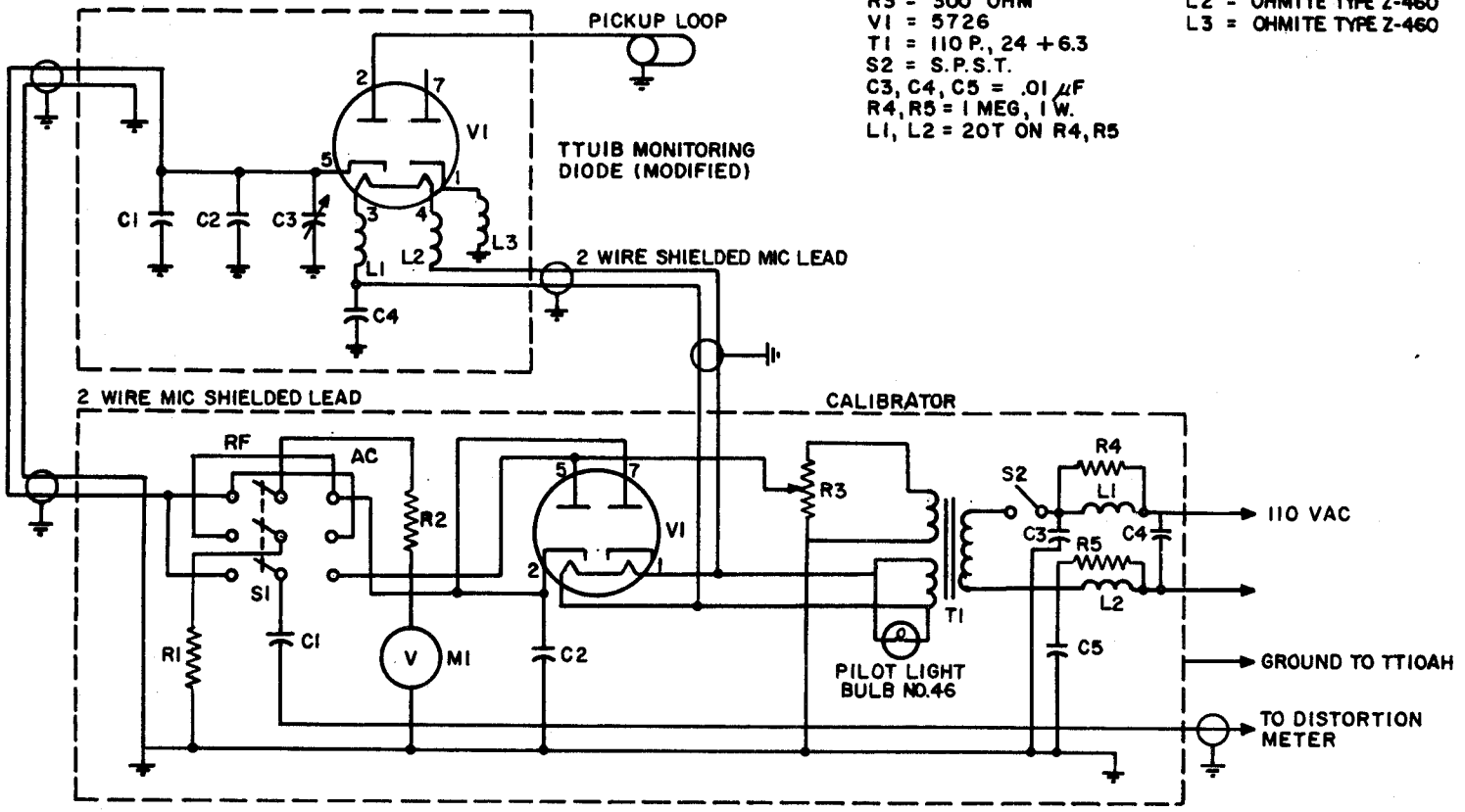
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CALIBRATOR

R1, R2 = 1 MEG. 1/10% WW
S1 = D.T.3 P
M1 = 50V (50 μA)
C1 = 4 μF, 600 VOLTS
C2 = 2 μF, 600 VOLTS
R3 = 300 OHM
V1 = 5726
T1 = 110 P., 24 + 6.3
S2 = S.P.S.T.
C3, C4, C5 = .01 μF
R4, R5 = 1 MEG, 1 W.
L1, L2 = 20T ON R4, R5

MONITORING DIODE

C1, C4 = 270 pF
C2 = .0001 μF
C3 = TRIMMER
V1 = 5726
L1 = OHMITE TYPE Z-460
L2 = OHMITE TYPE Z-460
L3 = OHMITE TYPE Z-460



NOTE:- MAKE ALL GROUNDS AT SAME POINT IN EACH CHASSIS.

FIGURE 2 Calibrator circuit

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